

Bridge Day 11 INSTRUCTOR KEY

While Loops, Update Order, and Termination Evidence

Learning sequence: Predict - Trace - Explain - Test - Interpret

Monday, July 20, 2026 • 20 points • target 18 minutes

Name		UIN		Section	
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Assessment boundary and evidence source

Contract: 07a04826c975

Cumulative foundations: input conversion, conditions, while, update order, termination

Scaffold level: complete bidirectional while-loop program

Evidence source: the matching v5 workbook readiness/evidence pages and VIP activities 18.19, 18.21, 18.22.

Show intermediate state for trace credit. Program credit requires one coherent solution, a stated assumption when the prompt leaves one implicit, and at least one test whose expected result can distinguish correct from incorrect logic.

Item 1. Trace a while loop that crosses its target. - 5 points

```
current = 2
target = 7
steps = 0

while current < target:
    current = current + 2
    steps = steps + 1
    print(current, steps)
```

Question: Give the complete pass ledger and explain why the final value can exceed target.

Model response: Passes store and print 4 1, 6 2, and 8 3. The condition is checked before each update; 6 is still below 7, so the next +2 update crosses the target. The loop stops only after current becomes 8.

Item 2. Diagnose a non-progressing loop. - 4 points

```
value = 3

while value < 10:
    print(value)
    value + 2
```

Question: Explain why the loop does not terminate, give the minimal repair, and list repaired output.

Model response: value + 2 computes a result but does not store it, so value remains 3 and the condition stays True. Repair with value = value + 2. Repaired output is 3, 5, 7, and 9 on separate lines.

Complete program - 11 points

- Read start and target as integers. Valid floors are 1 through 10 inclusive.
- Print invalid if either floor is outside the interval and do not run a loop.
- Print already there if the two valid floors are equal.
- Otherwise use a while loop to move exactly one floor per pass toward target and print each newly reached floor.
- After movement, print Steps: followed by the number of loop passes.

Program workspace

```
Model program
current = int(input())
target = int(input())

if current < 1 or current > 10 or target < 1 or target > 10:
    print("invalid")
elif current == target:
    print("already there")
else:
    steps = 0
    while current != target:
        if current < target:
            current = current + 1
        else:
            current = current - 1
        steps = steps + 1
        print(current)
    print(f"Steps: {steps}")
```

Test input, expected result, and why this test matters

Reasoning and test evidence Validation and equality are handled before repetition. The loop condition means movement is unfinished, and exactly one directional branch changes current by one each pass. Inputs 2 and 5 print 3, 4, 5, then Steps: 3. Inputs 5 and 2 print 4, 3, 2, then Steps: 3. This protects both update directions.

Scoring guidance: 2 validation/equality, 2 correct while condition, 3 directional update/progress, 2 output placement/count, 2 bidirectional tests.